# Measures of Center & Spread

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Mean and median**

Find the mean and median of the data set: $4, 7, 7, 9, 13$.

Mean $= \frac{4+7+7+9+13}{5} = \frac{40}{5} = 8$. Median (middle value of the ordered list): $7$.

**2** **[Foundational]** • **Target: Mean and median**

A sixth data point of $40$ is added to the set $4, 7, 7, 9, 13$. Recompute the mean and median, and explain which measure changed more.

New mean $= \frac{40+40}{6} = \frac{80}{6} \approx 13.33$. New median (ordered $4, 7, 7, 9, 13, 40$, average of the two middle values $7$ and $9$): $8$. The mean changed drastically (from $8$ to $13.33$) while the median barely moved (from $7$ to $8$), since the mean is sensitive to the extreme outlier $40$ while the median is resistant to it.

**3** **[Foundational]** • **Target: Range, IQR, and quartiles**

Find the five-number summary (min, $Q_1$, median, $Q_3$, max) of the ordered data: $2, 5, 6, 9, 11, 14, 15, 18, 20$.

Min $= 2$. Median (middle of 9 values): $11$. Lower half $2, 5, 6, 9$: $Q_1 = \frac{5+6}{2} = 5.5$. Upper half $14, 15, 18, 20$: $Q_3 = \frac{15+18}{2} = 16.5$. Max $= 20$.

**4** **[Foundational]** • **Target: Range, IQR, and quartiles**

Using the five-number summary from the previous problem, find the IQR and determine whether $20$ would be flagged as an outlier using the $1.5 \times$ IQR rule.

IQR $= Q_3 - Q_1 = 16.5 - 5.5 = 11$. Upper fence $= Q_3 + 1.5(11) = 16.5 + 16.5 = 33$ (using the $1.5 \times$ IQR rule). Since $20 < 33$, it is not flagged as an outlier.

**5** **[Foundational]** • **Target: Standard deviation and variance**

Find the standard deviation of the data set $2, 4, 6, 8$ (treat as a population).

Mean $= 5$. Deviations: $-3, -1, 1, 3$. Squared deviations: $9, 1, 1, 9$, sum $= 20$. Variance $= \frac{20}{4} = 5$. Standard deviation $= \sqrt{5} \approx 2.24$.

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**6** [Foundational] • Target: Standard deviation and variance

Explain why the standard deviation is always non-negative, and why it equals $0$ only in one specific case.

The standard deviation is the square root of an average of squared deviations, and squares are always non-negative, so their average and its square root are also non-negative. It equals $0$ only when every data value is identical (no spread at all), since that is the only case where every deviation from the mean is $0$.

**7** [Foundational] • Target: Visualizing spread with a normal-shaped distribution

The graph shows a bell-shaped distribution of exam scores with mean $75$ and standard deviation $8$.

**a** Estimate the range of scores that fall within one standard deviation of the mean. One standard deviation: $75 \pm 8$, so from $67$ to $83$.

**b** Estimate the range of scores that fall within two standard deviations of the mean. Two standard deviations: $75 \pm 16$, so from $59$ to $91$.

(a) One standard deviation: $75 \pm 8$, so from $67$ to $83$.; (b) Two standard deviations: $75 \pm 16$, so from $59$ to $91$.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Comparing distributions using measures of spread

Class A has a mean test score of $80$ with a standard deviation of $4$. Class B has a mean of $80$ with a standard deviation of $12$. Compare the two classes' consistency, and explain what a small vs. large standard deviation indicates.

Both classes have the same average performance (mean $80$), but Class A's scores are tightly clustered around the mean (small spread, standard deviation $4$), meaning most students scored similarly, while Class B's scores are much more spread out (standard deviation $12$), meaning there is far more variability — some students scored well above or below the average.

**9** [Intermediate] • Target: Resistant vs. non-resistant statistics

Explain why the median and IQR are described as "resistant" measures, while the mean and standard deviation are not, using the outlier example from earlier in this worksheet as a reference.

A resistant measure is one that is not strongly affected by extreme values or outliers. The median and IQR depend only on the position (ranking) of data values, not their exact size, so a single extreme value has little effect on them. The mean and standard deviation, in contrast, use every data value's actual size in their calculation, so one extreme outlier (like the $40$ added earlier) can shift them substantially.

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**10** **[Intermediate]** • **Target: Choosing appropriate summary statistics**

A data set of home prices in a neighborhood is strongly skewed right due to a few very expensive mansions. Recommend which pair of summary statistics (mean/standard deviation or median/IQR) should be reported, and justify your choice.

The median and IQR should be reported, since they are resistant to the effect of the few extremely expensive mansions. Reporting the mean would give a misleadingly high sense of a "typical" home price, since the mean is pulled upward by the small number of very high values.

**11** **[Intermediate]** • **Target: Choosing appropriate summary statistics**

A quality-control process measures the diameter of manufactured bolts, which are expected to be very consistent (low variability) and roughly symmetric. Recommend which pair of summary statistics is more appropriate here, and explain why.

The mean and standard deviation are appropriate here, since the data is roughly symmetric with no extreme outliers expected — in this case, the mean and standard deviation use all the data efficiently and are the standard choice when a distribution is not skewed.

**12** **[Intermediate]** • **Target: Computing summary statistics from a frequency table**

A survey of families records number of children per household: 0 children (8 families), 1 child (15 families), 2 children (20 families), 3 children (7 families). Find the mean number of children per household.

Total families: $8 + 15 + 20 + 7 = 50$. Total children: $0(8) + 1(15) + 2(20) + 3(7) = 0 + 15 + 40 + 21 = 76$. Mean $= \frac{76}{50} = 1.52$ children per household.

**13** **[Intermediate]** • **Target: Computing summary statistics from a frequency table**

A small business records weekly sales (in thousands): $10, 12, 12, 14, 17$. Find the mean, and then find the variance and standard deviation (treat as a population).

Mean $= \frac{10 + 12 + 12 + 14 + 17}{5} = \frac{65}{5} = 13$. Deviations: $-3, -1, -1, 1, 4$. Squared: $9, 1, 1, 1, 16$, sum $= 28$. Variance $= \frac{28}{5} = 5.6$. Standard deviation $= \sqrt{5.6} \approx 2.37$.

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**14** **[Intermediate]** • **Target: Computing summary statistics from a frequency table**

Two data sets both represent quiz scores out of 20 for two different classes of the same size, and both have the same mean of 15. Class A has a standard deviation of 1.2, while Class B has a standard deviation of 4.5. Without doing any further calculation, describe what a dotplot of each class's scores would likely look like, and explain how you can tell just from the two standard deviations.

Class A's dotplot would show scores tightly clustered close to 15 (most students scoring similarly, in a narrow band), since a small standard deviation of 1.2 indicates little spread around the mean. Class B's dotplot would show scores much more spread out around 15, with a substantial number of students scoring noticeably above or below the mean, since a standard deviation of 4.5 indicates far greater variability.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Computing summary statistics from a frequency table** • **Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A coach records the vertical jump heights (in inches) of 7 athletes: $22, 24, 25, 26, 26, 28, 41$. Explain why the value $41$ would likely be flagged as an outlier, and describe, in general terms, the two-step process for using the $1.5 \times$ IQR rule to formally check whether a value is an outlier.

The value $41$ is far larger than the rest of the data (which clusters between 22 and 28), so it looks like a clear outlier by visual inspection. Formally, the $1.5 \times$ IQR rule works in two steps: first compute the IQR ( $Q_3 - Q_1$) from the five-number summary, then compute the lower fence ($Q_1 - 1.5 \times$ IQR) and upper fence ($Q_3 + 1.5 \times$ IQR); any value falling below the lower fence or above the upper fence is classified as an outlier.

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**16** **[Challenging]** • **Target: Computing summary statistics from a frequency table** • **Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A teacher records the number of minutes 8 students spent on homework last night: $30, 45, 50, 55, 60, 60, 65, 200$. The value $200$ comes from a student who mistakenly reported total weekly homework time instead of just last night. Explain, step by step, how this single incorrect data entry would affect the reported mean versus the reported median, and explain which measure a teacher should trust more when summarizing typical homework time for this data set, given the known data-entry error.

The mean would be pulled dramatically upward by the erroneous $200$ value, since the mean uses every data point's exact size — it would suggest students spend far more time on homework than they actually do. The median, which only depends on the middle-ranked value(s), would be far less affected by this single extreme (and erroneous) value. Given the known data-entry error, the teacher should trust the median as a more accurate reflection of typical homework time, since it resists distortion from this kind of outli.

**17** **[Challenging]** • **Target: Mean and median** • **Cross-Lesson: 3.03 The Normal Distribution**

Level 2 · MediumThe heights of adult males in a region follow a normal distribution with mean $\mu = 70$ inches and standard deviation $\sigma = 3$ inches. What is the standardized z-score of a male who is 76 inches tall, and what proportion of adult males are taller than 76 inches?

Compute the z-score: $z = \frac{x - \mu}{\sigma} = \frac{76 - 70}{3} = \frac{6}{3} = \mathbf{2.00}$. By the Empirical Rule (68-95-99.7%), approximately 95% of data lies within 2 standard deviations of the mean, leaving 5% split between both tails. The upper tail is: $P(Z > 2.00) = \frac{1 - 0.95}{2} = 0.025 = \mathbf{2.5\%}$ (or 0.0228 using exact normal tables).

**18** **[Challenging]** • **Target: Range, IQR, and quartiles** • **Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

The commute times for employees at a large tech company have a strongly right-skewed distribution with mean $\mu = 38$ minutes and standard deviation $\sigma = 15$ minutes. A random sample of 49 employees is selected. (a) Describe the shape, mean, and standard error of the sampling distribution of the sample mean $\bar{x}$. (b) Calculate $P(\bar{x} > 42 \text{ minutes})$.

(a) Sampling Distribution Characteristics:Shape: By the Central Limit Theorem, since $n = 49 \geq 30$, the sampling distribution of $\bar{x}$ is approximately normal, despite the right skewness of the population.Mean: $\mu_{\bar{x}} = \mu = \mathbf{38}$ minutes.Standard Error: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{49}} = \frac{15}{7} \approx \mathbf{2.143}$ minutes.(b) Probability Calculation: Standardize: $z = \frac{42 - 38}{2.143} = \frac{4}{2.143} \approx 1.87$. .

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**19** [Challenging] • Target: Standard deviation and variance • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

Level 3 · AdvancedAn assembly line fills cereal boxes with a mean weight $\mu = 450$ grams and standard deviation $\sigma = 18$ grams. A random sample of $n = 36$ boxes is selected. What is the mean and standard error of the sample mean weight $\bar{x}$, and what is the probability that $\bar{x}$ exceeds 455 grams?

Sampling distribution parameters: $\mu_{\bar{x}} = \mu = \mathbf{450}$ g, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{18}{\sqrt{36}} = \frac{18}{6} = \mathbf{3}$ g. By the Central Limit Theorem ($n = 36 \geq 30$), $\bar{x}$ is approximately normally distributed: $z = \frac{455 - 450}{3} = \frac{5}{3} \approx 1.67$. Evaluating tail probability: $P(\bar{x} > 455) = P(Z > 1.67) = 1 - 0.9525 = \mathbf{0.0475}$ (4.75%).

**20** [Challenging] • Target: Visualizing spread with a normal-shaped distribution • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A set of daily high temperatures in Celsius has mean $\bar{x} = 25.0^\circ\text{C}$ and standard deviation $s = 4.0^\circ\text{C}$. Convert the temperatures to Fahrenheit using $F = 1.8C + 32$. Find the new mean and standard deviation in Fahrenheit.

Mean: Affected by both multiplication and addition: $\bar{x}_F = 1.8(25.0) + 32 = 45.0 + 32 = \mathbf{77.0}^\circ\text{F}$.
Standard Deviation: Affected ONLY by multiplication, not by the addition of 32:
$s_F = 1.8(s_C) = 1.8(4.0) = \mathbf{7.2}^\circ\text{F}$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Mean and median

Find the mean of the data set: $17, 28, 12, 6, 13$.

**a.** $\frac{81}{5}$ **b.** $\frac{71}{5}$

**c.** $19$ **d.** $\frac{76}{5}$ ✓

Mean $= \frac{17 + 28 + 12 + 6 + 13}{5} = \frac{76}{5}$.

**22** [Foundational] • Target: Range, IQR, and quartiles

Find the median of the data set: $16, 15, 5, 25, 29, 12$.

**a.** $\frac{31}{2}$ ✓ **b.** $17$

**c.** $\frac{33}{2}$ **d.** $\frac{37}{2}$

Ordered: $5, 12, 15, 16, 25, 29$. Median (average of the two middle values $15$ and $16$): $\frac{15 + 16}{2} = \frac{31}{2}$.

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**23** [Foundational] • Target: Standard deviation and variance

Find the IQR of the ordered data set: $4, 7, 20, 29, 43, 50, 51, 54$.

**a.** 36

**b.** $\frac{101}{2}$

**c.** 38

**d.** 37 ✓

Lower half $4, 7, 20, 29$: $Q_1 = \frac{7+20}{2} = \frac{27}{2}$. Upper half $43, 50, 51, 54$: $Q_3 = \frac{50+51}{2} = \frac{101}{2}$. IQR $= Q_3 - Q_1 = 37$.

**24** [Foundational] • Target: Visualizing spread with a normal-shaped distribution

A data set has $Q_1 = 22$ and $Q_3 = 43$. Using the $1.5 \times$ IQR rule, find the upper fence used to detect outliers.

**a.** $\frac{147}{2}$

**b.** 64

**c.** $\frac{149}{2}$ ✓

**d.** $\frac{151}{2}$

IQR $= Q_3 - Q_1 = 43 - 22 = 21$. Upper fence $= Q_3 + 1.5 \times \text{IQR} = 43 + 1.5(21) = \frac{149}{2}$.

**25** [Foundational] • Target: Comparing distributions using measures of spread

Find the standard deviation of the data set $14, 10, 13, 23$ (treat as a population), to two decimal places.

**a.** 5.15

**b.** 4.85 ✓

**c.** 4.55

**d.** 5.45

Mean $= 15$. Deviations: $-1, -5, -2, 8$. Squared deviations: $1, 25, 4, 64$, sum $= 94$. Variance $= \frac{94}{4} = \frac{47}{2}$. Standard deviation $= \sqrt{\frac{47}{2}} \approx 4.85$.

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**26** [Foundational] • Target: Resistant vs. non-resistant statistics

A machine dispenses a liquid into containers. The amount of liquid dispensed, in milliliters, follows a continuous uniform distribution over the interval $[95, 105]$. A quality control inspector randomly selects 5 containers and measures the amount of liquid in each. Let $X_i$ be the amount of liquid in the $i^{th}$ container, for $i = 1, 2, 3, 4, 5$. The company has implemented a new calibration process that, on average, increases the mean amount of liquid dispensed by $\delta$ milliliters, but also increases the variance by $\sigma^2$. The inspector is specifically looking for evidence that the calibration process has *decreased* the variability of the dispensed liquid while maintaining a satisfactory average fill. If the current calibration process yields a mean fill of $\mu_0 = 100$ ml and a standard deviation of $\sigma_0 = 2$ ml, and the new process is hypothesized to have a mean fill of $\mu_1 = 100 + \delta$ ml and a standard deviation of $\sigma_1 = 2 - \sigma_{dev}$ ml, where $\delta > 0$ and $\sigma_{dev} > 0$ are unknown positive constants, what is the minimum value of $\sigma_{dev}$ such that the probability of falsely concluding that the new process has *increased* variability (i.e., Type I error) is at most $0.01$, assuming the mean fill remains at $100$ ml (i.e., $\delta = 0$)?

**a.** The probability cannot be determined without knowing the specific value of $\delta$.

**b.** Approximately $1.01$

**c.** **Approximately** $1.37$ ✓

**d.** Approximately $2.00$

Let $X$ be the amount of liquid dispensed by the machine. $X$ follows a continuous uniform distribution $U(95, 105)$. The mean is $\mu = \frac{95+105}{2} = 100$ ml. The variance is $\sigma^2 = \frac{(105-95)^2}{12} = \frac{100}{12} = \frac{25}{3}$. The standard deviation is $\sigma = \sqrt{\frac{25}{3}} = \frac{5}{\sqrt{3}} \approx 2.887$ ml. The problem statement implies the initial standard deviation is $\sigma_0 = 2$ ml, which contradicts the calculated standard deviation for a uniform distribution from 95 to 10.

**27** [Foundational] • Target: Choosing appropriate summary statistics

The scores on a standardized test have a symmetric, bell-shaped distribution with mean 500 and standard deviation 100. Approximately what percentage of test takers scored between 400 and 600?

**a.** 50%

**b.** **68%** ✓

**c.** 95%

**d.** 99.7%

By the Empirical Rule (68-95-99.7 rule), approximately 68% of data in a normal distribution lies within 1 standard deviation of the mean ($500 \pm 100$).

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**28** [Foundational] • Target: Computing summary statistics from a frequency table

A pharmaceutical company is conducting a clinical trial to test the efficacy of a new drug designed to lower blood pressure. They recruit 200 participants and randomly assign them to either the treatment group (receiving the new drug) or the control group (receiving a placebo). After 12 weeks, they measure the reduction in systolic blood pressure for each participant. The company wants to perform a hypothesis test to determine if there is a significant difference in blood pressure reduction between the two groups. The trial is designed with 20 participants in the treatment group and 180 participants in the control group. If the standard deviation of blood pressure reduction in the treatment group is $\sigma_1 = 15$ mmHg and in the control group is $\sigma_2 = 12$ mmHg, and they are planning to use a two-sample t-test, what is the calculated pooled standard deviation $s_p$ if equal variances are assumed, given the following summary statistics for blood pressure reduction (in mmHg): $\bar{x}_1 = 10$ mmHg for the treatment group and $\bar{x}_2 = 7$ mmHg for the control group?

**a.** $\sqrt{\frac{15^2 + 12^2}{2}}$

**b.** $\sqrt{\frac{20(15^2) + 180(12^2)}{200}}$ ✓

**c.** $\frac{20 \times 15 + 180 \times 12}{200}$

**d.** $\sqrt{\frac{19(15^2) + 179(12^2)}{198}}$

The problem asks for the pooled standard deviation $s_p$ when assuming equal variances for a two-sample t-test. The formula for the pooled variance $s_p^2$ is given by: $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$. In this problem, we have: - Sample size for treatment group, $n_1 = 20$ - Sample standard deviation for treatment group, $s_1 = 15$ mmHg - Sample size for control group, $n_2 = 180$ - Sample standard deviation for control group, $s_2 = 12$ mmHg - Total degrees of.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** **[Intermediate]** • **Target: Mean and median**

A biostatistician is developing a new statistical test for comparing the proportion of successful treatments across three independent groups. The test involves calculating a test statistic that follows a chi-square distribution under the null hypothesis. The null hypothesis involves comparing the proportions of a binary outcome (e.g., recovery or no recovery) across these three groups. The data is presented in a contingency table. If the contingency table has 3 rows representing the treatment groups and 5 columns representing the outcome categories (e.g., various stages of recovery), what is the expected value of the test statistic under the null hypothesis, assuming the expected cell counts are all greater than or equal to 5, and given that the overall proportion of success across all groups is $p = 0.7$? Assume for the purpose of this problem that the true proportions for each group are known and lead to a specific deviation from the null, such that the sum of the squared differences between observed and expected counts, divided by the expected counts, is calculated. Consider a hypothetical scenario where the observed proportions for the three groups lead to expected cell counts that perfectly align with the grand proportion $p = 0.7$ for the 'success' column and $1 - p = 0.3$ for the 'no success' column. If the total sample size across all groups is $N = 150$, and the degrees of freedom for the chi-square test are $df$, what is the value of $\frac{1}{df}\sum\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$ where $O_{ij}$ are the observed counts and $E_{ij}$ are the expected counts under the null hypothesis that group proportions are equal?

**a.** 0.7

**b.** **1.0** ✓

**c.** 1.5

**d.** 2.1

The problem describes a chi-square test for comparing proportions across multiple groups. The degrees of freedom (df) for a chi-square test of independence or homogeneity in a $r \times c$ contingency table is given by $(r - 1)(c - 1)$. In this scenario, we have $r = 3$ (treatment groups) and $c = 2$ (outcome categories: success and no success). Thus, $df = (3 - 1)(2 - 1) = 2 \times 1 = 2$. The question asks for the value of $\frac{1}{df}\sum\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$. The term.

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**30** **[Intermediate]** • **Target: Range, IQR, and quartiles**

A marketing research firm is designing an experiment to test the effectiveness of three different advertising slogans (Slogan A, Slogan B, Slogan C) on product purchase intent. They recruit 240 participants and randomly assign them to one of the three slogan groups. After exposure to their assigned slogan, participants rate their purchase intent on a scale of 1 to 10. The experiment is designed as a completely randomized design. The data collected will be analyzed using ANOVA to compare the mean purchase intent scores across the three groups. In order to determine the minimum number of participants needed in each group for a power analysis, the researchers need to estimate the effect size. A pilot study suggests that the standard deviation of purchase intent scores within each group is approximately $\sigma = 1.8$. They aim for a minimum of $n = 80$ participants per group to achieve the desired statistical power. If the total number of participants available for this experiment were $N = 240$ and they decided to allocate the participants into $k = 3$ treatment groups, with an equal number of participants in each group, what would be the number of participants per group, $n$? Additionally, if the experimenters decided to use a blocked design instead, where participants are first blocked by age (Young, Middle-aged, Senior) and then randomly assigned slogans within each block, how many blocks would be needed?

**a.** **80 participants per group; 3 blocks ✓**

**b.** 60 participants per group; 2 blocks

**c.** 80 participants per group; 2 blocks

**d.** 60 participants per group; 3 blocks

The problem asks for two pieces of information: the number of participants per group in a completely randomized design and the number of blocks in a blocked design. Part 1: Participants per group in a completely randomized design. Total number of participants $N = 240$. Number of treatment groups $k = 3$. Since the participants are to be equally divided among the groups, the number of participants per group $n$ is calculated as: $n = \frac{N}{k} = \frac{240}{3} = 80$. Part 2: Number of blocks in a blocked design. T.

**31** **[Intermediate]** • **Target: Standard deviation and variance**

A population has $\mu = 141$ and $\sigma = 30$. For a random sample of size $n = 36$, find $P(\bar{x} < 143)$ using the sampling distribution of the sample mean.

**a.** 0.7554

**b.** 0.9554

**c.** 0.3554

**d.** **0.6554 ✓**

$\sigma_{\bar{x}} = \sigma/\sqrt{n} = 5$. $z = \frac{143 - 141}{5} = 0.4$. $P(\bar{x} < 143) = 0.6554$.

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**32** [Intermediate] • Target: Visualizing spread with a normal-shaped distribution

A population proportion is $p = 0.38$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of $\hat{p}$.

**a.** 0.058

**b.** **0.028 ✓**

**c.** 0.018

**d.** 0.068

$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.38 \times 0.62}{300}} = 0.028.$

**33** [Intermediate] • Target: Comparing distributions using measures of spread

A population has $\sigma = 30$. For random samples of size $n = 9$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error).

**a.** 13

**b.** 6

**c.** 12

**d.** **10 ✓**

$\sigma_{\bar{x}} = \sigma/\sqrt{n} = 30/\sqrt{9} = 30/3 = 10.$

**34** [Intermediate] • Target: Resistant vs. non-resistant statistics

A distribution has mean $\mu = 71$ and standard deviation $\sigma = 8$. What is the z-score for $x = 55$?

**a.** -1.5

**b.** **-2 ✓**

**c.** -0.5

**d.** -2.5

$z = \frac{x - \mu}{\sigma} = \frac{55 - 71}{8} = -2.$

**35** [Intermediate] • Target: Choosing appropriate summary statistics

Which parameter describes spread in a Normal distribution?

**a.** Mean

**b.** Median

**c.** **Standard deviation ✓**

**d.** Variance only

Standard deviation σ measures spread.

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**36** [Intermediate] • Target: Computing summary statistics from a frequency table

The box plot below shows the distribution of scores for a class on a recent statistics quiz. The five-number summary is given by minimum, first quartile (Q1), median, third quartile (Q3), and maximum. If the interquartile range (IQR) is 25, and the median score is 70, find the value of Q1.

**a.** **45** ✓

**b.** 60

**c.** 75

**d.** 95

The interquartile range (IQR) is defined as the difference between the third quartile (Q3) and the first quartile (Q1), i.e., $\text{IQR} = Q_3 - Q_1$. We are given that the IQR is 25. The median of the dataset is given as 70. In a box plot, the median is represented by the line inside the box. Therefore, $Q_2 = 70$. The box plot also indicates that the third quartile (Q3) is 70. This is because the right edge of the box aligns with the median line, indicating $Q_3 = 70$. Now we can use the IQR formula to find Q1.

**37** [Intermediate] • Target: Mean and median

A data set of 20 distinct positive integers has a mean of 50 and a median of 45. If the largest number in the data set is increased by 100, which of the following statements must be true? I. The mean increases. II. The median increases. III. The standard deviation increases.

**a.** I only

**b.** I and II only

**c.** **I and III only** ✓

**d.** I, II, and III

Increasing the single largest value increases the total sum, so the mean increases (I is true). It does not affect the middle numbers, so the median does not change (II is false). It spreads the data further from the mean, increasing the standard deviation (III is true). Thus I and III only.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Range, IQR, and quartiles • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A pharmaceutical company is testing a new drug to reduce blood pressure. They conduct a clinical trial with $n = 100$ participants. The sample mean reduction in systolic blood pressure is $\bar{x} = 15$ mmHg with a sample standard deviation of $s = 8$ mmHg. The company plans to construct a 95% confidence interval for the true mean reduction in systolic blood pressure. If the margin of error for this confidence interval is $M = 2$ mmHg, what would be the required sample size, $n'$, to achieve a margin of error of $M' = 1$ mmHg with the same confidence level and an estimated population standard deviation equal to the sample standard deviation?

**a.** 200

**b.** 300

**c.** 400

**d.** **1600 ✓**

The formula for the margin of error (ME) of a confidence interval for a population mean is given by $\text{ME} = z^* \frac{\sigma}{\sqrt{n}}$, where $z^*$ is the critical value for the desired confidence level, $\sigma$ is the population standard deviation, and $n$ is the sample size. In the initial scenario, we have $n = 100$, $s \approx \sigma = 8$ mmHg, and $M = 2$ mmHg. For a 95% confidence interval, the critical value $z^*$ is approximately 1.96. Using the margin of error formula.

**39** [Challenging] • Target: Standard deviation and variance • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A software company develops a new encryption algorithm. The algorithm's security depends on a sequence of random integers generated by a custom pseudo-random number generator (PRNG). The PRNG is designed such that the probability of generating an integer in a specific range $[a, b]$ is proportional to the length of the range, $b - a$, for $0 \le a \le b \le 1$. Let $X$ be the first integer generated and $Y$ be the second integer generated by this PRNG, independently. The problem is to find the probability that the sum of the first two generated integers, $S = X + Y$, is less than 1. Assume the PRNG generates numbers from a continuous uniform distribution on $[0, 1]$.

**a.** 1/4

**b.** **1/3 ✓**

**c.** 1/2

**d.** 2/3

The problem states that the probability of generating an integer in a range $[a, b]$ is proportional to $b - a$ for $0 \le a \le b \le 1$. This describes a continuous uniform distribution on the interval $[0, 1]$. Therefore, the probability density function (PDF) for both $X$ and $Y$ is $f(x) = 1$ for $0 \le x \le 1$ and 0 otherwise. We are interested in the probability $P(X + Y < 1)$. Since $X$ and $Y$ are independent and both follow a $U(0, 1)$ distribution, their joint PDF is $f(x, y) = f(x)f(y) = 1$ for.

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**40** [Challenging] • Target: Visualizing spread with a normal-shaped distribution • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A quality control engineer is monitoring the number of defects produced by two different machines, Machine A and Machine B. Let $X$ be the number of defects produced by Machine A in a given hour, and $Y$ be the number of defects produced by Machine B in the same hour. The joint probability distribution of $X$ and $Y$ is given in the table below: | $P(X = x, Y = y)$ | $y = 0$ | $y = 1$ | $y = 2$ | | :----------: | :-----: | :-----: | :-----: | | $x = 0$ | 0.10 | 0.15 | 0.05 | | $x = 1$ | 0.20 | 0.30 | 0.10 | | $x = 2$ | 0.05 | 0.05 | 0.00 | What is the expected value of the total number of defects produced by both machines in a given hour, $E(X + Y)$?

**a.** 1.15

**b.** **1.30** ✓

**c.** 1.45

**d.** 1.60

To find the expected value of the total number of defects, $E(X + Y)$, we can use the property of linearity of expectation: $E(X + Y) = E(X) + E(Y)$. First, let's calculate the marginal probability distributions for $X$ and $Y$. For $X$: $P(X = 0) = 0.10 + 0.15 + 0.05 = 0.30$ $P(X = 1) = 0.20 + 0.30 + 0.10 = 0.60$ $P(X = 2) = 0.05 + 0.05 + 0.00 = 0.10$ Now, calculate $E(X)$: $E(X) = (0 \times 0.30) + (1 \times 0.60) + (2 \times 0.10) = 0 + 0.60 + 0.20 = 0.80$ For $Y$: $P(Y = 0) = 0.10 + 0.20 + 0.05 = 0.35$.

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**41** [Challenging] • Target: Comparing distributions using measures of spread • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A marketing research firm is analyzing the relationship between the preferred social media platform and age group of consumers. They collected data from a sample and summarized it in the two-way table below. The marginal distributions are also provided. The research team is interested in understanding how the distribution of preferred platform varies across age groups and is considering a chi-square test for independence. If the null hypothesis of independence is rejected, the observed counts will be significantly different from what would be expected under independence. Assume that a chi-square test for independence is to be performed on this data. Which of the following represents a potential challenge or insight derived from the structure of the data and the test's assumptions?

**a.** The number of degrees of freedom for the chi-square test can be determined solely from the total number of observations, irrespective of the table dimensions.

**b.** If the expected count in any cell is less than 5, a normal approximation to the binomial distribution for each cell count would be appropriate to proceed with the chi-square test.

**c.** **The calculation of expected counts under the null hypothesis of independence relies on the marginal proportions of the observed data.** ✓

**d.** A high chi-square statistic indicates that the observed counts are very close to the expected counts, suggesting a strong association between age group and preferred social media platform.

The chi-square test for independence assumes that the expected counts in each cell are sufficiently large, typically at least 5. Option A is incorrect because degrees of freedom depend on the number of rows and columns: df = (r-1)(c-1). Option B is incorrect; if expected counts are too small, alternative methods like Fisher's exact test or combining categories might be necessary, not a normal approximation for cell counts in this context. Option D is incorrect; a high chi-square statistic indicates that the observe.

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**42** [Challenging] • Target: Resistant vs. non-resistant statistics • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A researcher is investigating the relationship between two categorical variables, 'Exercise Frequency' and 'Dietary Preference', using a chi-square test of independence. The contingency table summarizing the observed frequencies has $3$ rows and $5$ columns. What is the number of degrees of freedom $\text{df}$ for this chi-square test?

**a.** 15

**b.** **12** ✓

**c.** 8

**d.** 6

The degrees of freedom $\text{df}$ for a chi-square test of independence are calculated using the formula $\text{df} = (r-1)(c-1)$, where $r$ is the number of rows and $c$ is the number of columns in the contingency table. In this case, $r = 3$ and $c = 5$. Therefore, $\text{df} = (3-1)(5-1) = (2)(4) = 8$.

**43** [Challenging] • Target: Choosing appropriate summary statistics • Cross-Lesson: 2.01 Sampling Methods & Bias

Standard deviation of Binomial(n,p) is:

**a.** $np$

**b.** $np(1-p)$

**c.** $\sqrt{np(1-p)}$ ✓

**d.** $\sqrt{np}$

Applying Unit 3 · Probability, Random Variables & Distributions principles, σ=√(np(1-p)). Therefore, the verified correct option is $\sqrt{np(1-p)}$.

**44** [Challenging] • Target: Computing summary statistics from a frequency table • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A variable follows a Normal distribution with $\mu = 78$ and $\sigma = 6$. Using the empirical rule, approximately what percent of values lie between 72 and 84?

**a.** 50%

**b.** 99.7%

**c.** **68%** ✓

**d.** 95%

72 and 84 are $\mu \pm 1\sigma$, so by the empirical rule approximately 68% of values fall in this range.

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**45** [Challenging] • Target: Mean and median • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

Two boxplots are compared: Group A has median 30 and IQR 17; Group B has median 43 and IQR 20. Which statement correctly compares their spread?

**a. Group B has greater variability (larger IQR) than Group A ✓**

**b.** The group with the higher median always has less spread

**c.** Group A and Group B have the same median

**d.** Both groups are perfectly symmetric

Group A IQR = 17, Group B IQR = 20. Group B has greater variability (larger IQR) than Group A.

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